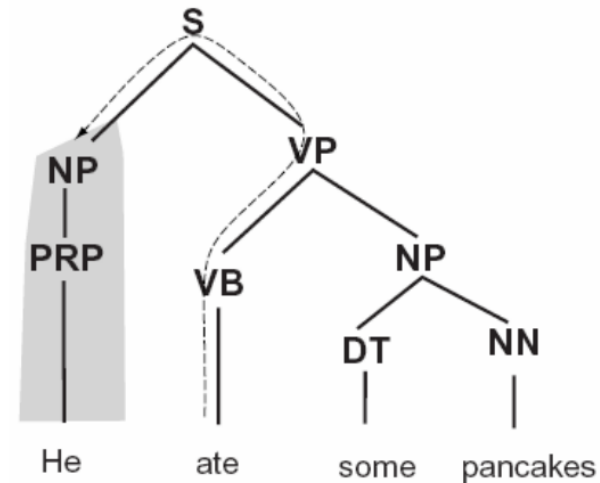


Distant Supervision for Relation Extraction

- Hand-labeled data with relation labels is expensive to produce
- However, available resources such as Wikipedia infoboxes, have a great many relations that are in structured format
 - Wikipedia articles contain sentences that express these relations
 - This huge amount of examples allows us to define rich features

Path Features

- Path features:
 - Encode the path between two nodes, through the direction of the edges and their labels
- Similar features exist for dependency parses
 - We will see a bit of this in Exercise 5



<i>Path</i>	<i>Description</i>
VB↑VP↓PP	PP argument/adjunct
VB↑VP↑S↓NP	subject
VB↑VP↓NP	object
VB↑VP↑VP↑S↓NP	subject (embedded VP)
VB↑VP↓ADVP	adverbial adjunct
NN↑NP↑NP↓PP	prepositional complement of noun

Examples of Rich Features

...Hubble was born in Marshfield...
...Einstein, born 1879, Ulm...
...Hubble's birthplace in Marshfield...



PER was born in LOC
PER, born *, LOC
PER's birthplace is LOC

American Airlines, a unit of AMR,
immediately matched the move,
spokesman **Tim Wagner** said



Constituent path	$NP \uparrow NP \uparrow S \uparrow S \downarrow NP$
Base phrase path	$NP \rightarrow NP \rightarrow PP \rightarrow NP \rightarrow VP \rightarrow NP \rightarrow NP$
Typed-dependency path	$Airlines \leftarrow_{subj} matched \leftarrow_{comp} said \rightarrow_{subj} Wagner$

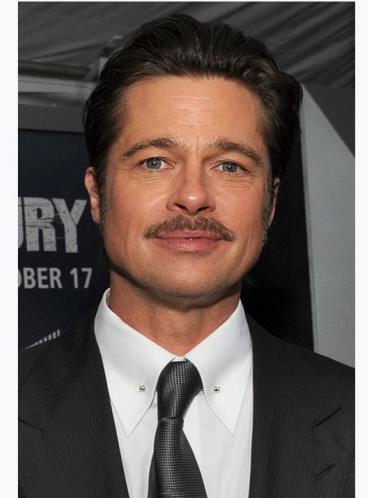
Schematized Algorithm for Distant Supervision

```
function DISTANT SUPERVISION(Database D, Text T) returns relation classifier C  
  
  foreach relation R  
    foreach tuple (e1, e2) of entities with relation R in D  
      sentences  $\leftarrow$  Sentences in T that contain e1 and e2  
      f  $\leftarrow$  Frequent features in sentences  
      observations  $\leftarrow$  observations + new training tuple (e1, e2, f, R)  
  C  $\leftarrow$  Train supervised classifier on observations  
  return C
```

Example: Distant Supervision

- From the infobox we learn that “Brad Pitt” and “Jennifer Aniston” were spouses
- The Wikipedia text states:
 - “In 2000, **he** married actress **Jennifer Aniston**; they divorced in 2005.”
 - “... playing a man with a grudge against Rachel Green, played by **Jennifer Aniston**, to whom **Pitt** was married at the time.”
 - “... Plan B Entertainment, a film production company **he** had founded two years earlier with **Jennifer Aniston** and Brad Grey”
- The first two examples are examples as to how spouse relation may be expressed in the text; the third example is noise

Brad Pitt



Pitt at the premiere of *Fury* in [Washington, D.C.](#), October 2014

Born	William Bradley Pitt December 18, 1963 (age 53) Shawnee, Oklahoma, U.S.
Occupation	Actor • producer
Years active	1987–present
Works	Filmography
Home town	Springfield, Missouri
Spouse(s)	Jennifer Aniston (m. 2000; div. 2005) Angelina Jolie (m. 2014; separated 2016)
Children	6
Relatives	Douglas Pitt (brother)